Test Results

Summary Tables

The averages in the tables below are from over the 5 minute test window.

Cloud

Test Run ID	Mpbs	Avg Latency (ms)	Avg no req/sec >10ms	Avg Flow Rate (per min)	Avg CPU (%)	Avg RAM (Mb)
Cloud_1	10	0.0609	0.126	18662	94.8	170
Cloud_2	50	0.124	0.738	62241	98.0	180
Cloud_3	100	0.210	2.26	86197	94.3	255
Cloud_4	500	0.187	2.14	92935	96.9	279

Edge

Test Run ID	Mpbs	Avg Latency (ms)	Avg no req/sec >10ms	Avg Flow Rate (per min)	Avg CPU (%)	Avg RAM (Mb)
Edge_1	10	0.0649	0.111	18153	87.7	188
Edge_2	50	0.103	0.566	62016	96.1	180
Edge_3	100	0.188	2.02	103596	94.8	208
Edge_4	500	0.192	2.13	98908	97.2	286

Full Output: MiniKube Cloud Environment

Test Cloud_1

Command

```
tcpreplay -i eth0 --mbps=10 --loop=0 /pcaps/bigflows.pcap
```

tcpreplay final stats

```
Actual: 1087796 packets (477004112 bytes) sent in 381.60 seconds
Rated: 1250003.7 Bps, 10.00 Mbps, 2850.60 pps
```

```
Flows: 40467 flows, 106.04 fps, 1086865 unique flow packets, 616 unique non-flow packets
Statistics for network device: eth0
    Successful packets:        1087795
    Failed packets:            0
    Truncated packets:         0
    Retried packets (ENOBUFS): 0
    Retried packets (EAGAIN):  0
```

Dashboard Graphs

Test Cloud_2

Command

```
tcpreplay -i eth0 --mbps=50 --loop=0 /pcaps/bigflows.pcap
```

tcpreplay final stats

```
Actual: 5062183 packets (2234001099 bytes) sent in 357.44 seconds
Rated: 6250003.0 Bps, 50.00 Mbps, 14162.32 pps
Flows: 40467 flows, 113.21 fps, 5057842 unique flow packets, 2801 unique non-flow packets
Statistics for network device: eth0
    Successful packets:        5062182
    Failed packets:            0
    Truncated packets:         0
    Retried packets (ENOBUFS): 0
    Retried packets (EAGAIN):  0
```

Dashboard Graphs

Test Cloud_3

Command

```
tcpreplay -i eth0 --mbps=100 --loop=0 /pcaps/bigflows.pcap
```

tcpreplay final stats

```
Actual: 11213151 packets (4952924919 bytes) sent in 396.23 seconds
Rated: 12500031.3 Bps, 100.00 Mbps, 28299.38 pps
Flows: 40467 flows, 102.12 fps, 11203478 unique flow packets, 6211 unique
non-flow packets
Statistics for network device: eth0
    Successful packets:        11213150
    Failed packets:            0
    Truncated packets:         0
    Retried packets (ENOBUFS): 0
    Retried packets (EAGAIN):  0
```

Dashboard Graphs

Test Cloud_4

Command

```
tcpreplay -i eth0 --mbps=500 --loop=0 /pcaps/bigflows.pcap
```

tcpreplay final stats

```
Actual: 57904481 packets (25576607583 bytes) sent in 409.22 seconds
Rated: 62500110.1 Bps, 500.00 Mbps, 141497.90 pps
Flows: 40467 flows, 98.88 fps, 57854634 unique flow packets, 31930 unique
non-flow packets
Statistics for network device: eth0
    Successful packets:        57904480
    Failed packets:            0
    Truncated packets:         0
    Retried packets (ENOBUFS): 0
    Retried packets (EAGAIN):  0
```

Dashboard Graphs

Full Output: k3s Edge Environment

Test Edge_1

Command

```
tcpreplay -i eth0 --mbps=10 --loop=0 /pcaps/bigflows.pcap
```

tcpreplay final stats

```
Actual: 1094922 packets (481223506 bytes) sent in 384.97 seconds
Rated: 1250000.0 Bps, 10.00 Mbps, 2844.10 pps
Flows: 40467 flows, 105.11 fps, 1093990 unique flow packets, 617 unique non-flow packets
Statistics for network device: eth0
    Successful packets:        1094921
    Failed packets:            0
    Truncated packets:         0
    Retried packets (ENOBUFS): 0
    Retried packets (EAGAIN):  0
```

Dashboard Graphs

Test Edge_2

Command

```
tcpreplay -i eth0 --mbps=50 --loop=0 /pcaps/bigflows.pcap
```

tcpreplay final stats

```
Actual: 5179221 packets (2291284163 bytes) sent in 366.60 seconds
Rated: 6250007.9 Bps, 50.00 Mbps, 14127.52 pps
Flows: 40467 flows, 110.38 fps, 5174735 unique flow packets, 2875 unique
non-flow packets
Statistics for network device: eth0
    Successful packets:        5179220
    Failed packets:            0
    Truncated packets:         0
    Retried packets (ENOBUFS): 0
    Retried packets (EAGAIN):  0
```

Dashboard Graphs

Test Edge_3

Command

```
tcpreplay -i eth0 --mbps=100 --loop=0 /pcaps/bigflows.pcap
```

tcpreplay final stats

```
Actual: 10261426 packets (4530451507 bytes) sent in 362.43 seconds
Rated: 12500004.1 Bps, 100.00 Mbps, 28312.38 pps
Flows: 40467 flows, 111.65 fps, 10252596 unique flow packets, 5647 unique
non-flow packets
Statistics for network device: eth0
    Successful packets:        10261425
    Failed packets:            0
    Truncated packets:         0
    Retried packets (ENOBUFS): 0
    Retried packets (EAGAIN):  0
```

Dashboard Graphs

Test Edge_4

Command

```
tcpreplay -i eth0 --mbps=500 --loop=0 /pcaps/bigflows.pcap
```

tcpreplay final stats

```
Actual: 53085046 packets (23445745413 bytes) sent in 375.13 seconds
Rated: 62500154.3 Bps, 500.00 Mbps, 141510.68 pps
Flows: 40467 flows, 107.87 fps, 53039354 unique flow packets, 29261 unique
non-flow packets
Statistics for network device: eth0
    Successful packets:        53085045
    Failed packets:            0
    Truncated packets:         0
    Retried packets (ENOBUFS): 0
    Retried packets (EAGAIN):  0
```

Dashboard Graphs

xApp latency (ms)
0.2025
0.2
0.1975
0.195
0.1925
0.19
19:26:00
19:27:00
19:28:00
19:29:00
19:30:00
Average
Average xApp latency (ms)
0.192
xApp Flow Rate (per min)
105000
100000
95000
90000
19:26:00
19:27:00
19:28:00
19:29:00
19:30:00
Flow Rate
Average Flow Rate (per min)
98908
xApp Requests >10ms/s (req/sec)
3
2
1
19:26:00
19:27:00
19:28:00
19:29:00
19:30:00
Requests >10ms/s
Average No Requests >10ms per second
2.13
Container RAM Usage (Mb)
288
286
284
282
280
19:26:00
19:27:00
19:28:00
19:29:00
19:30:00
RAM Usage
Average RAM Usage
286
CPU Usage (%)
100
95
90
85
80
75
19:26:00
19:27:00
19:28:00
19:29:00
19:30:00
CPU Usage
Average CPU Usage (%)
97.2